Computing the Heavens: An Introduction to Computational Astrophysics

Instructor:
Ian Johnson

Recap of Last Week (with slides)

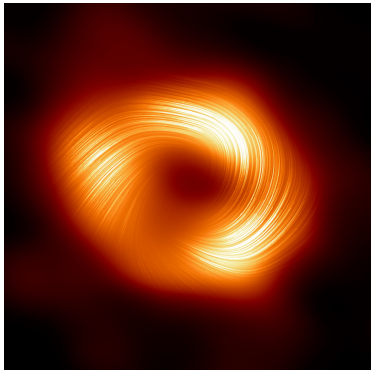
Who am I?

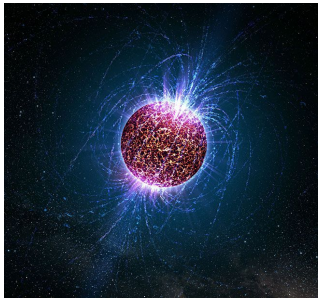

San Bernardino
Southern California
Santa Barbara
Ventura
Los Angeles
Orange
Riverside
Imperial
San Diego

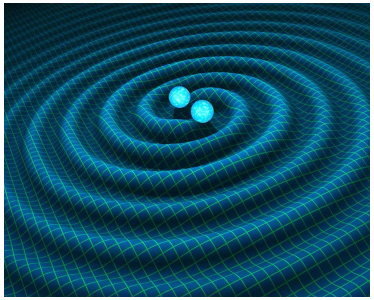


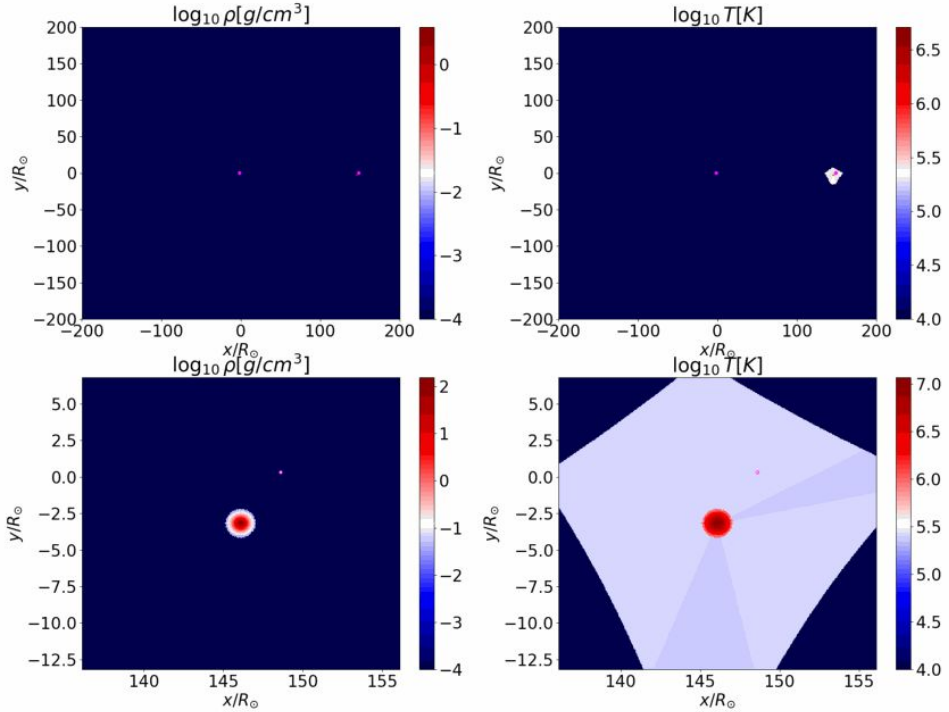

PHYSICS

What Do I Study?

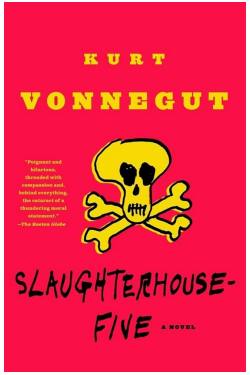






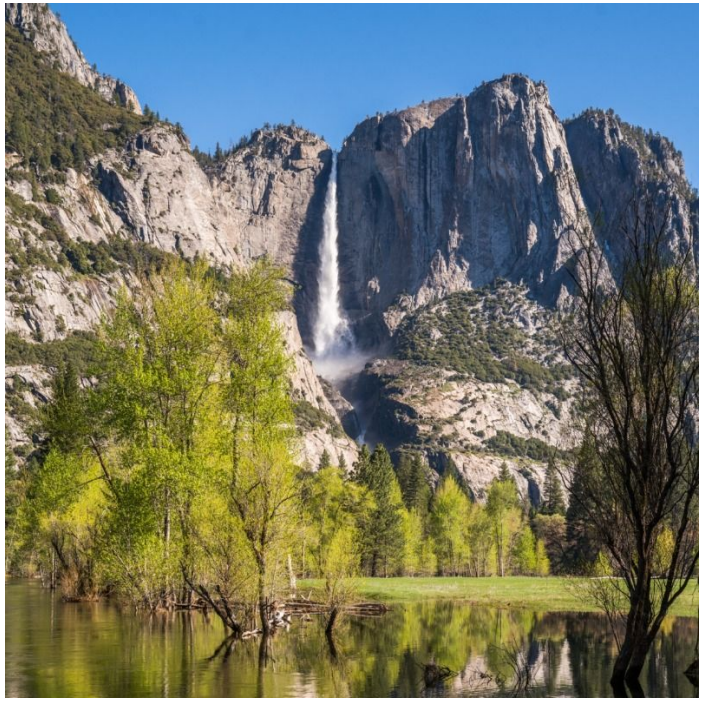


Outside of Work...

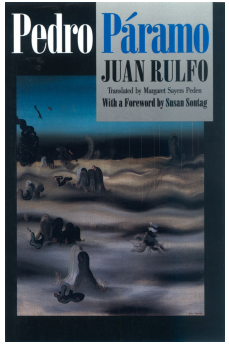












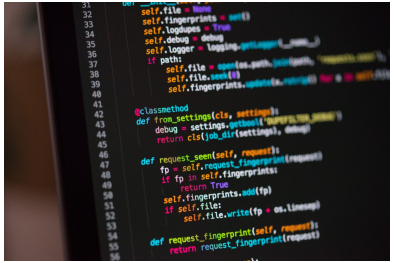
What is Computational Astrophysics?

What is Computational Astrophysics?

- Physics
- Astronomy
- Software





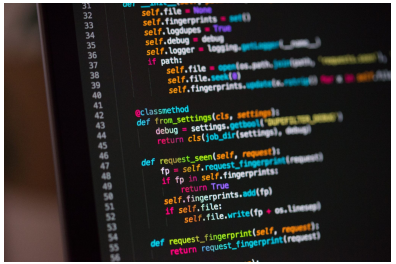


What is Computational Astrophysics?

- Physics
  - Math
  - Model Design
- Astronomy
  - Chemistry
  - Optics
- Software
  - Coding
  - Hardware







$$f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$$

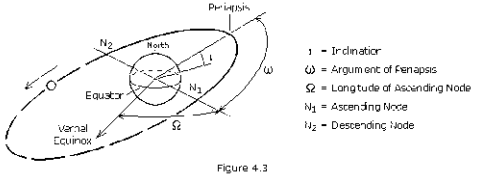
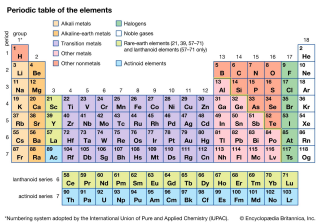
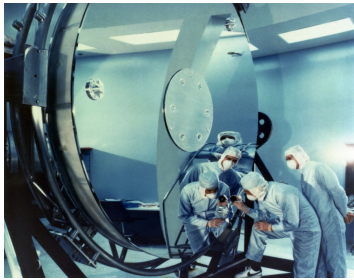


Figure 4.3







And More!

An Astronomy Overview:

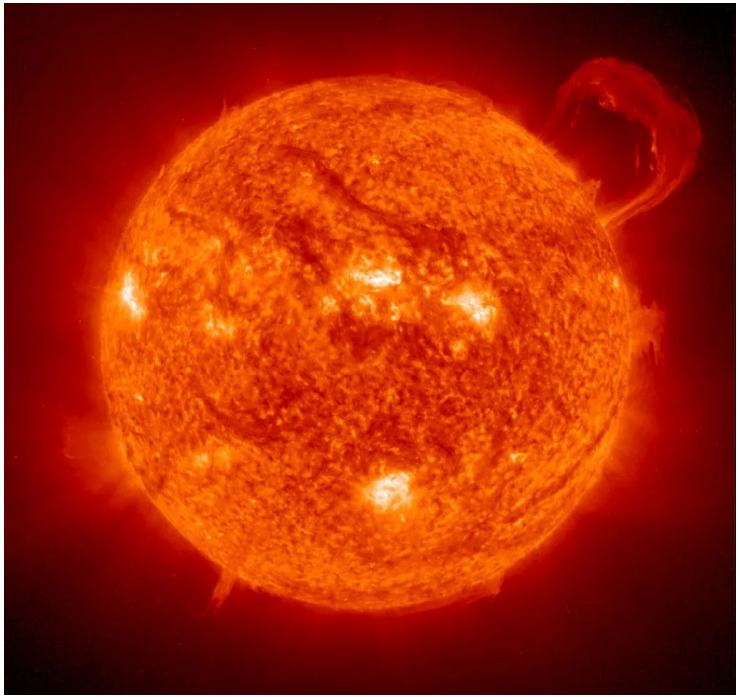
Basics of Astronomy

Stars:

The Building Blocks of Astronomy

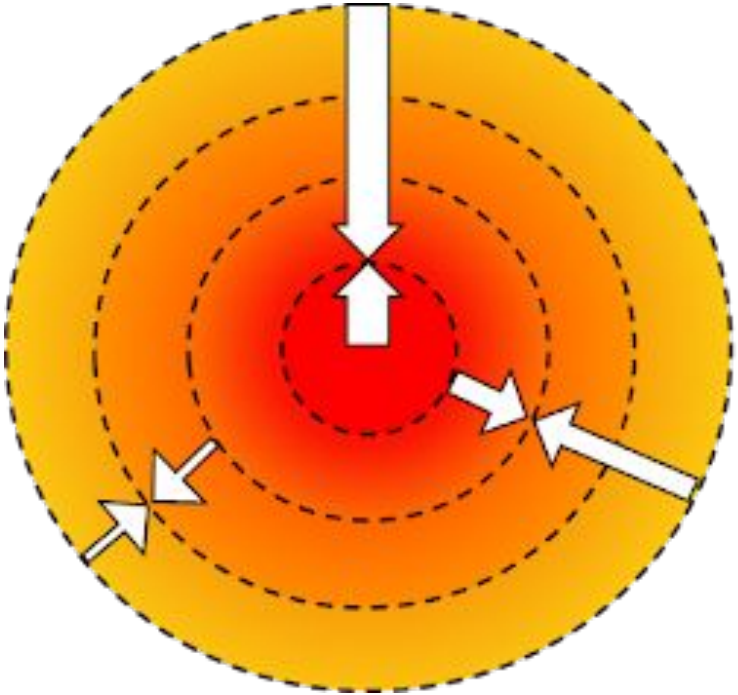
What Are Stars Made of?

- 75% Hydrogen
- 25% Helium
- Trace other elements
- Plasma and a bit of gas



Our Sun—the basis of most stellar astrophysics

Hydrostatic Equilibrium



Hydrostatic equation

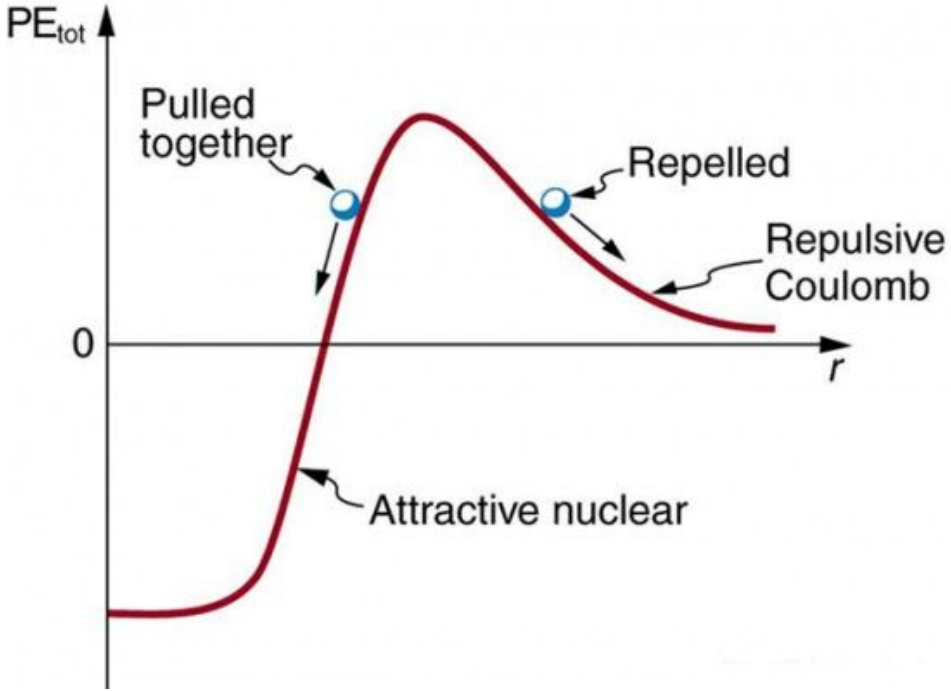
$$\rho_r \frac{d^2 r}{dt^2} = -\frac{dP_r}{dr} - G\frac{M_r \rho_r}{r^2}$$

Hydrostatic equilibrium

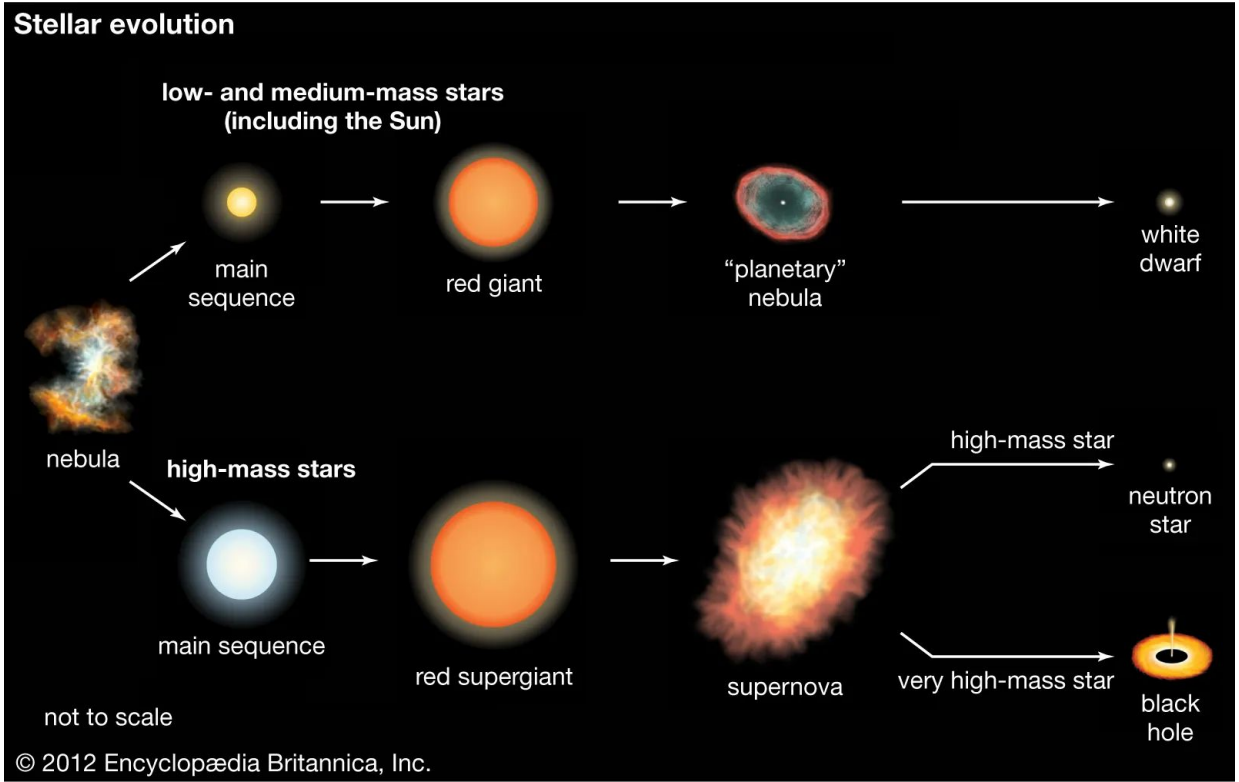
$$\frac{dP_r}{dr} = -G\frac{M_r \rho_r}{r^2}$$

Fusion

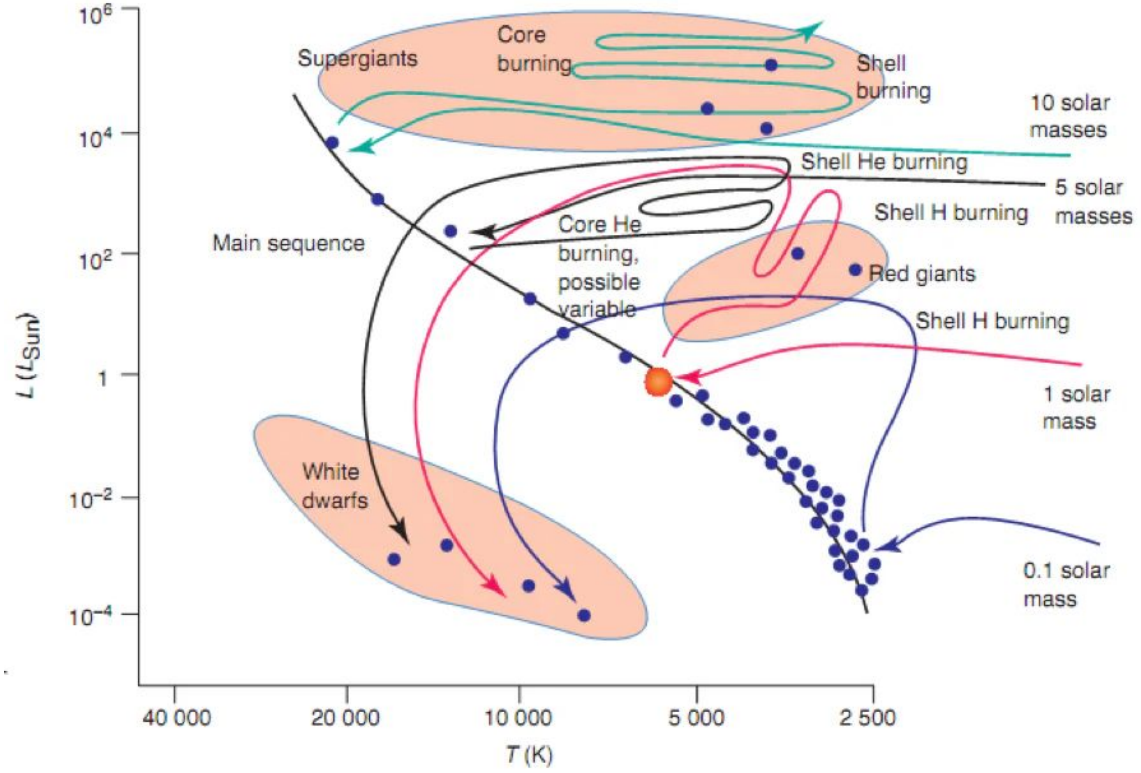
- Fusion Burns Up to Iron
- Fusion releases heat which builds pressure
- Stars are the creators of almost all elements



Stellar Evolution



Main Sequence



Galaxies:

The size of the universe

Galaxies are Star Groups

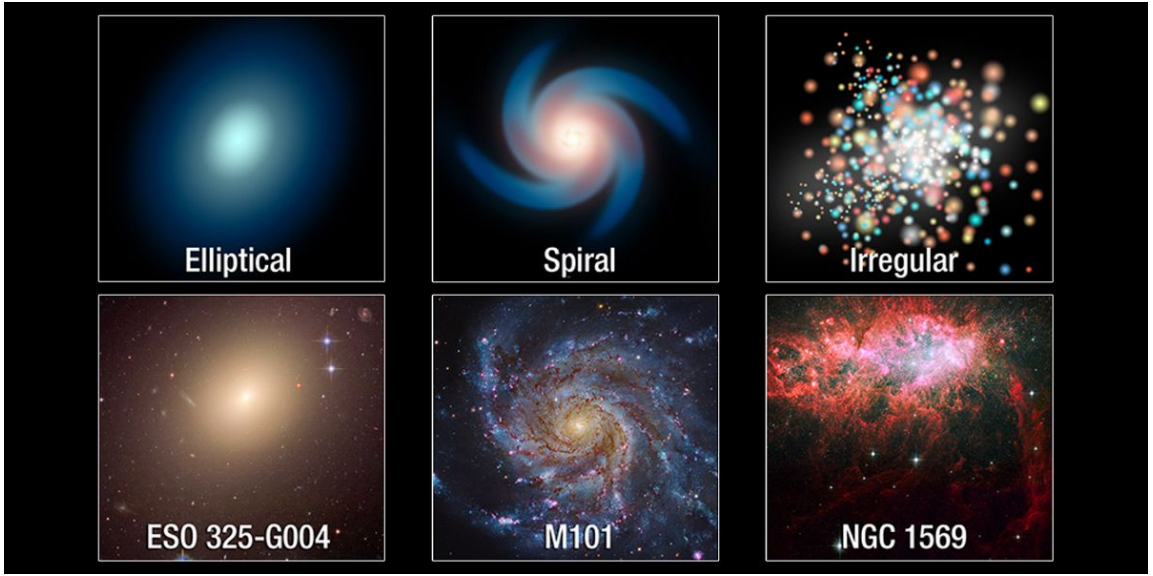
- 10^5 to 10^{13} solar masses of stars
- Most of the mass is dark matter
- ~100,000 light years across (~10 kpc)
- Have dwarf galaxies
- Centered on a supermassive black hole



Types of Galaxies

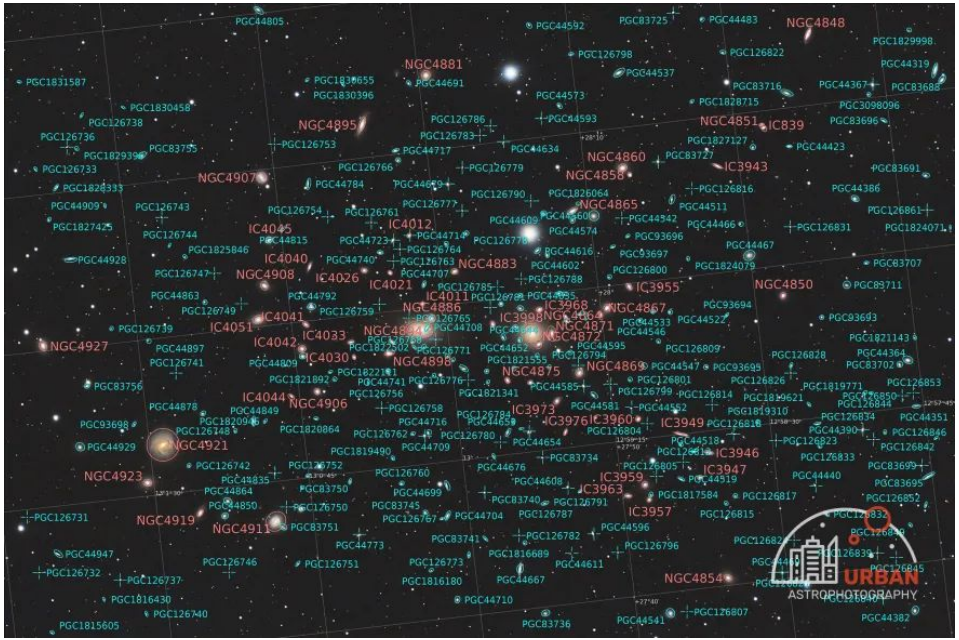
- Types:
  - Spiral
  - Elliptical
  - Irregular
  - S0
- Spirals:
  - Star Forming
  - Young + Brighter
  - Disk structure
- Ellipticals:
  - Old + Dimmer
  - Spherical

What causes the difference?



Things Can Get Bigger

- Groups have a few galaxies
- Galaxy clusters have 1000s of galaxies
- In between is made of very hot gas (ten million Kelvin!)
- But mostly dark matter…

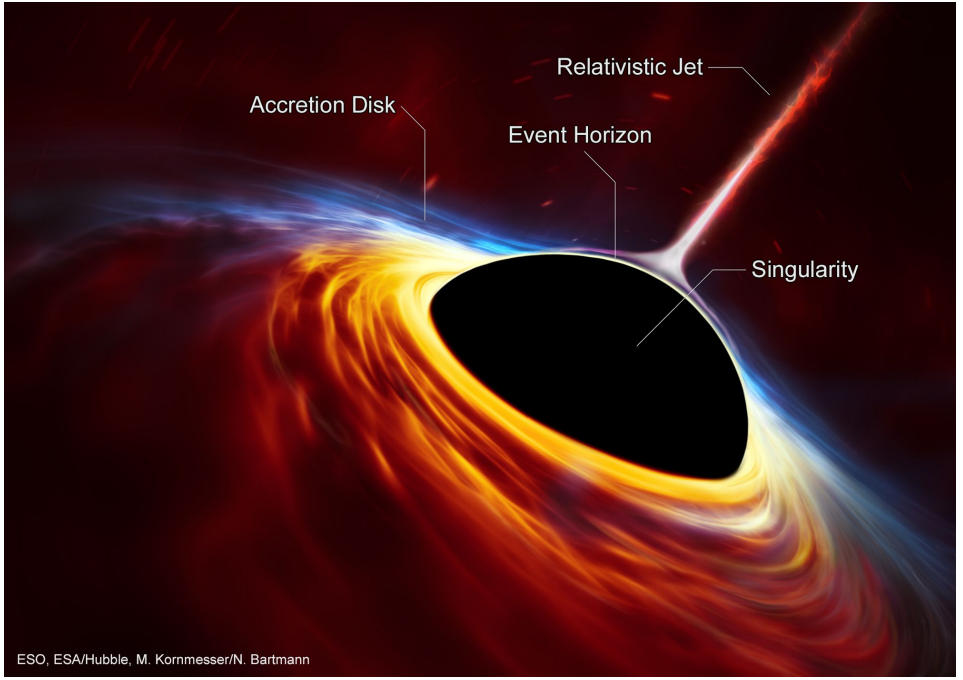


Compact Objects:

The Oddballs of Astronomy

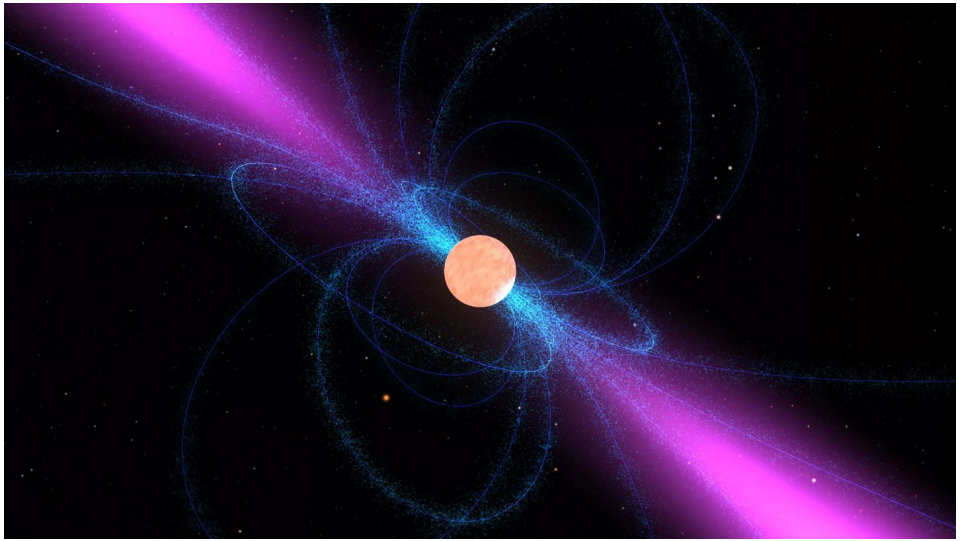
Black Holes

- Relativity says nothing can exceed the speed of light
- Black holes formed via star death
  - Or during the early universe
- We can only see their accretion
- Can be observed by gravitational effects



Neutron Stars

- Just shy of being a black hole
- The whole star is like an atom
- Hosts up to 1.6 billion Tesla magnetic fields
- Used as “standard clocks” (error of 1.7×10^{-17} s)



Cosmology:
The History of the Universe

Exoplanets:

And the search for life

Presentations!

Presentations

- Introduce yourselves again
- Make 3-5 slides to speak for 5-10 minutes
- Teach me about either:
  - The basics of your topic
  - The specifics of a part of your topic
- Compile work on one computer using other computers to do background research
- We will work for ~30 minutes
- Don’t worry about knowing everything and please ask me clarifying questions

Group	Member A	Member B	Member C	Member D
# 1	Isabella	Thalia	Tiana	Olivia
# 2	Samantha	Hirushi	Katerina	Jadyn
# 3	Ella	Maddy	Ava	

Stony Brook University

Women in Science & Engineering (WISE)

We are offering CPR Training from the University Police Department!

When? 4-5 PM on Thursday, Nov 21st, 2024

Where? Stony Brook Main Campus Student Union Building Multipurpose Room

Cost? Class is free, but certification is $20.00 payable by check at the class

There are 20 spaces available, so first-come first-serve. We will contact you regarding you status.

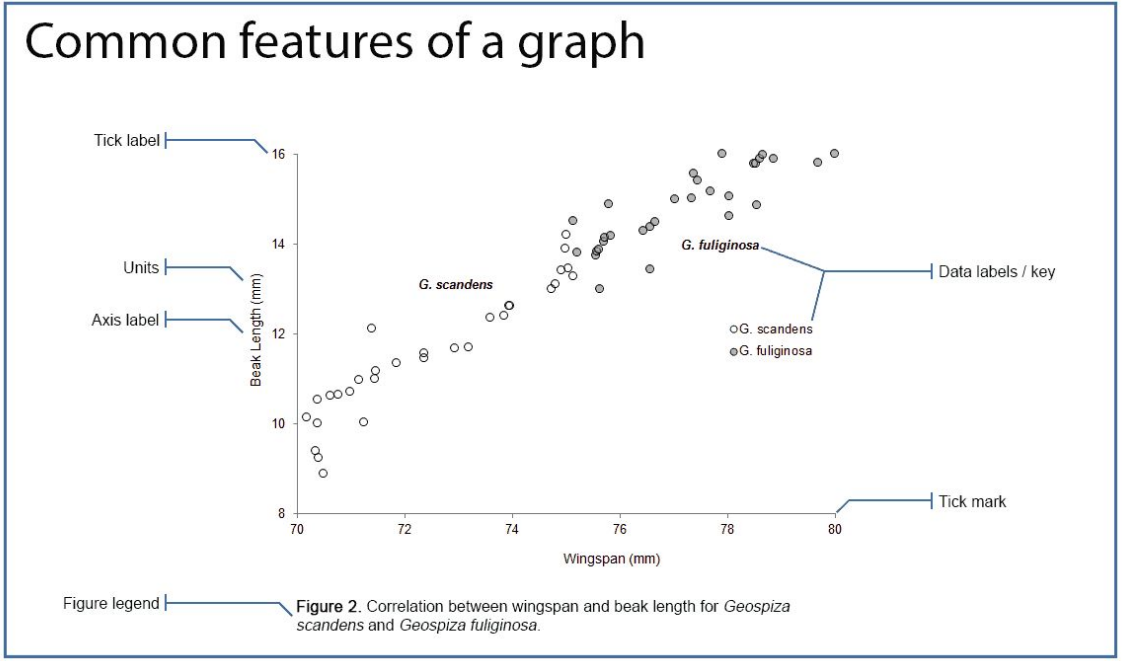
SCAN ME

Register via the QR code or this link: https://forms.gle/CXGnE6dgQiCS8wSF6

Group Picture!
(Now or Later?)

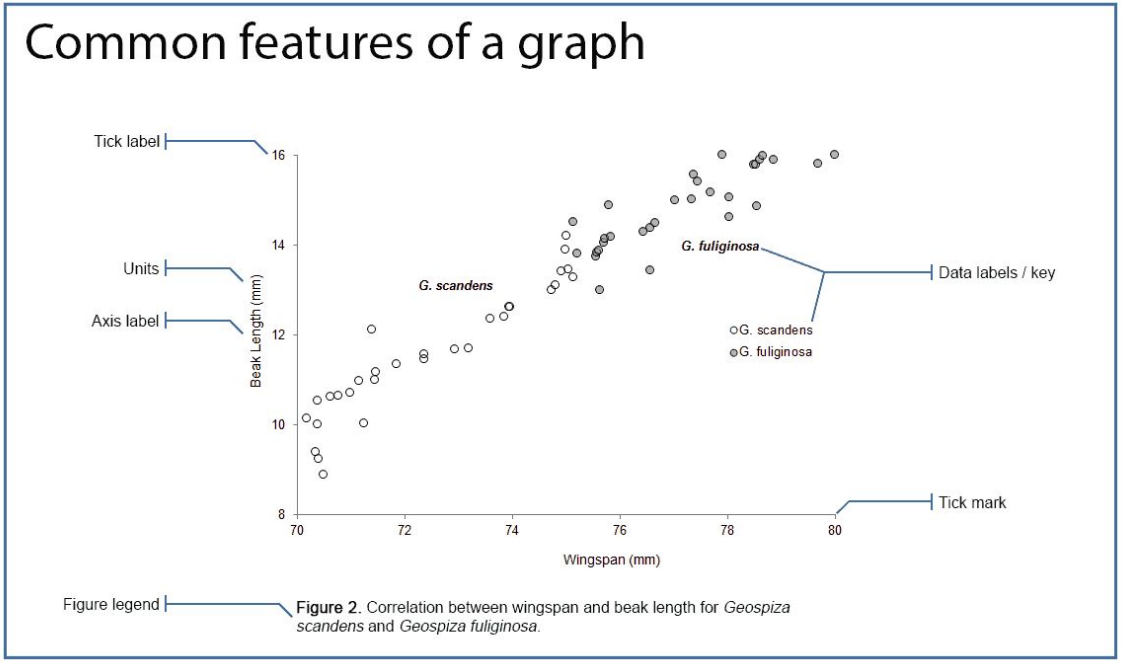
How to Read Plots in Astro

Anatomy of a Plot



Anatomy of a Plot

- Title
- Axes
  - Labels
  - Units
  - Scale
- Legend
- Caption

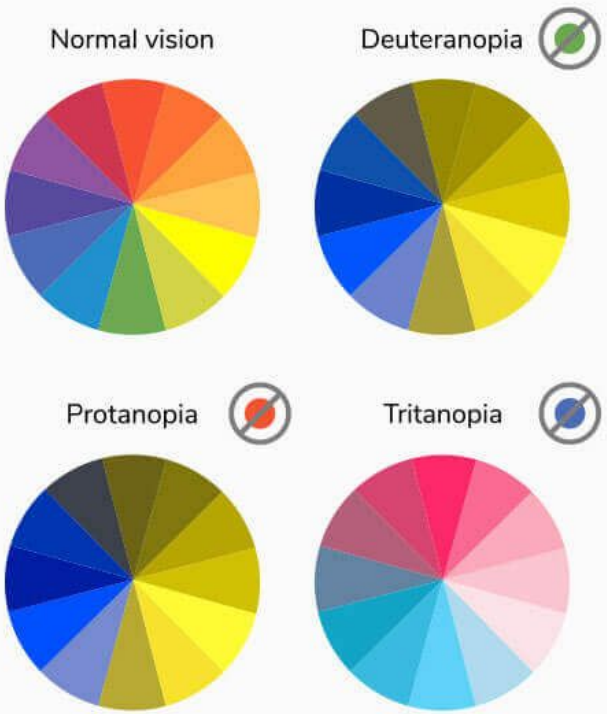


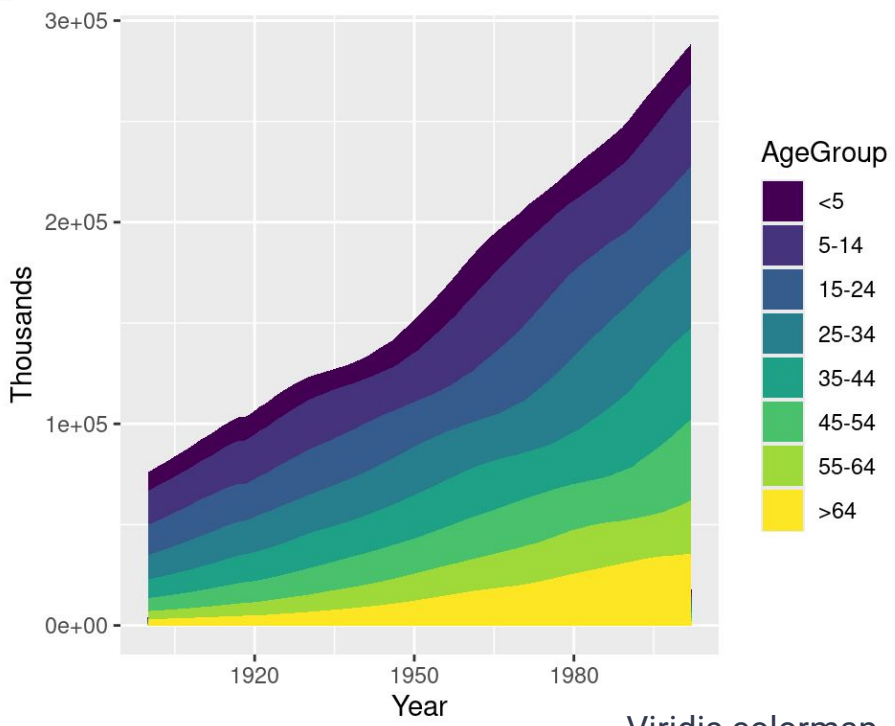
Axioms of Plot Design

Plots should…

- Speak for themselves
- Be easy to read for everyone
- Maintain scientific rigor

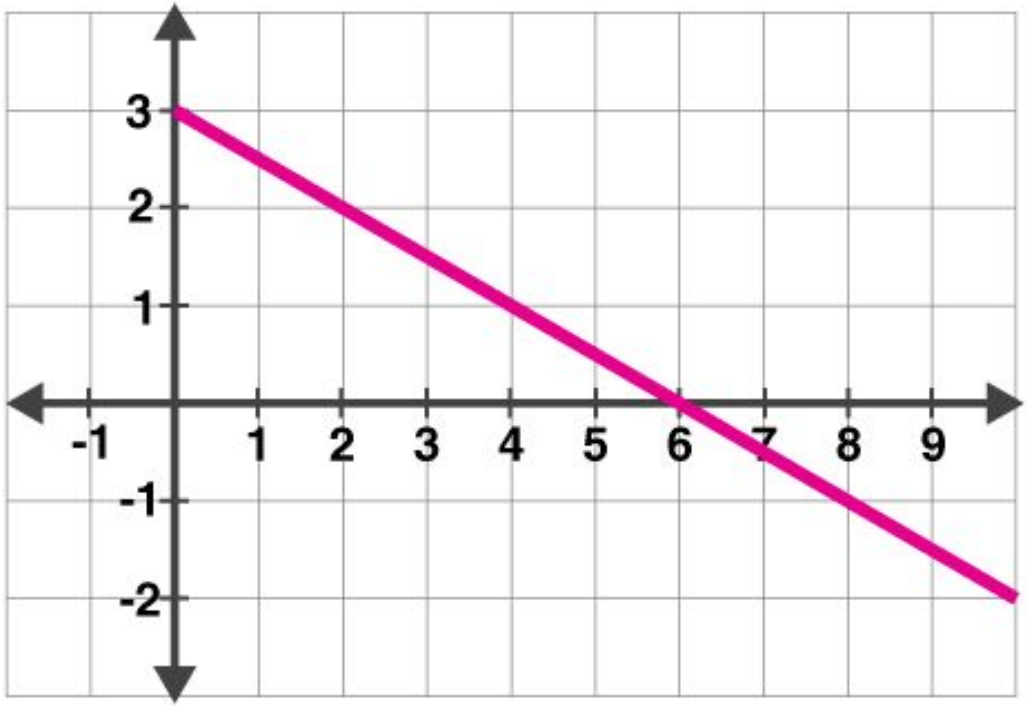
Color Blindness





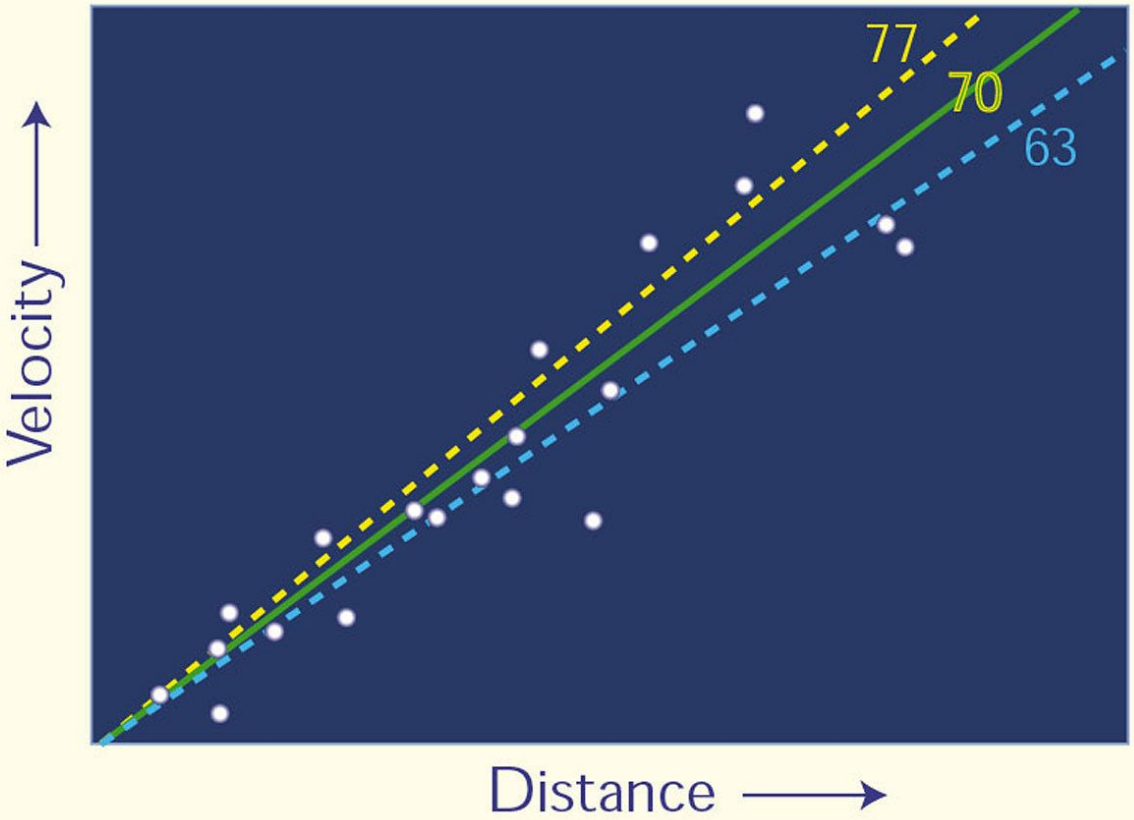
Viridis colormap

Let’s Judge Some Plots!

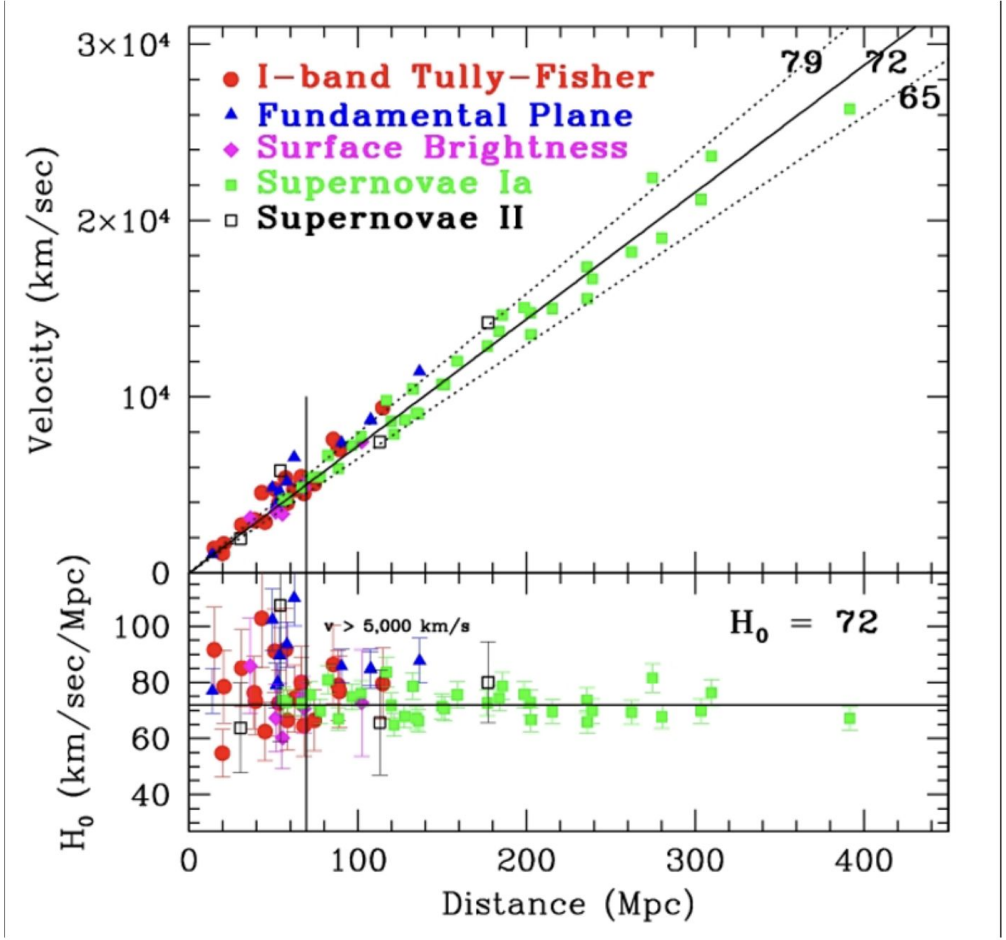


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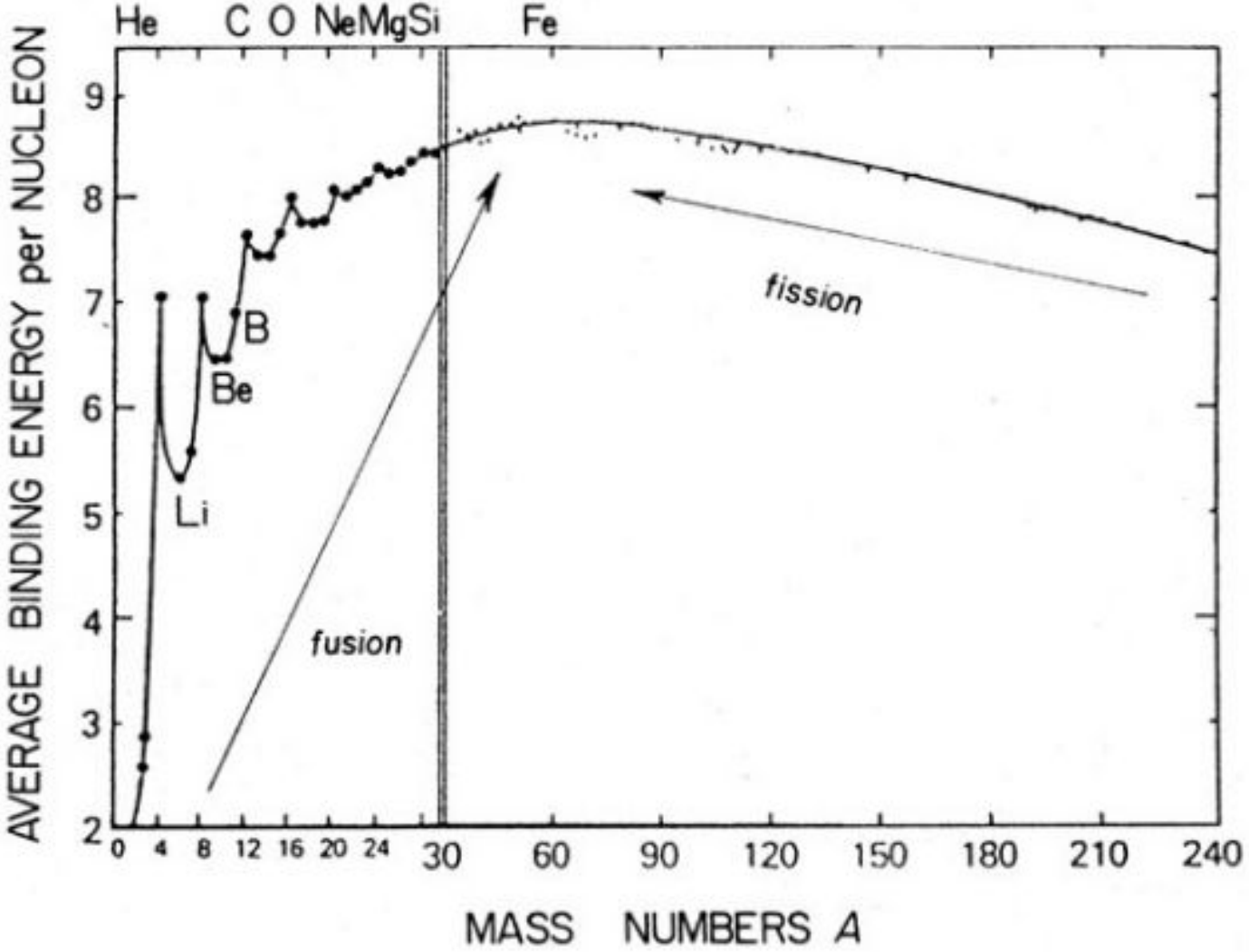
Hubble Diagram for Cepheids

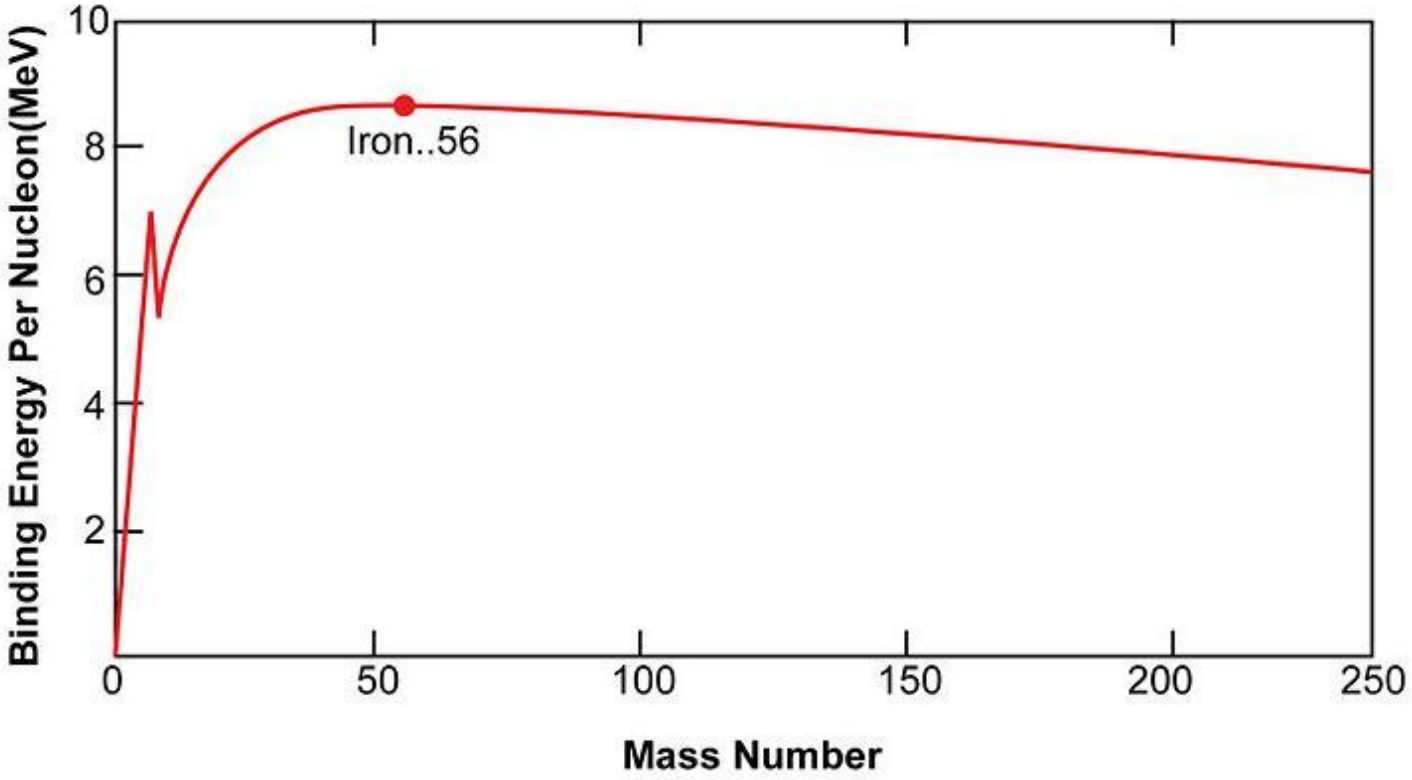


Caption: Hubble diagram showing the final results from the HST Key Project (Freedman et al. 2001) In the top panel velocity is plotted versus distance for the five secondary distance methods labeled in the plot. The solid line is for a slope of 72, flanked by dotted lines indicating a 10% uncertainty. The bottom panel shows the residuals for a value of H0 = 72 ± 3 (stat) ±7 (sys) km s −1 Mpc −1 .

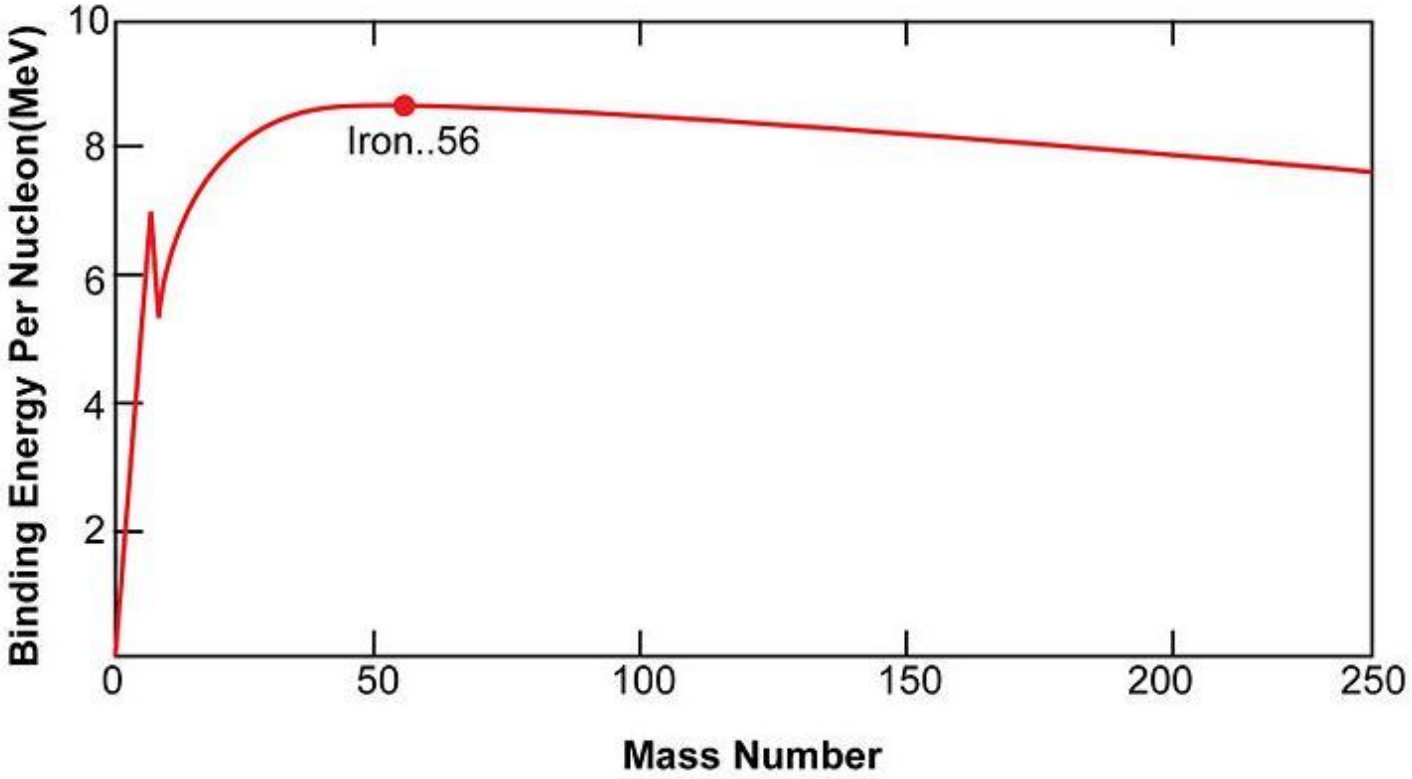


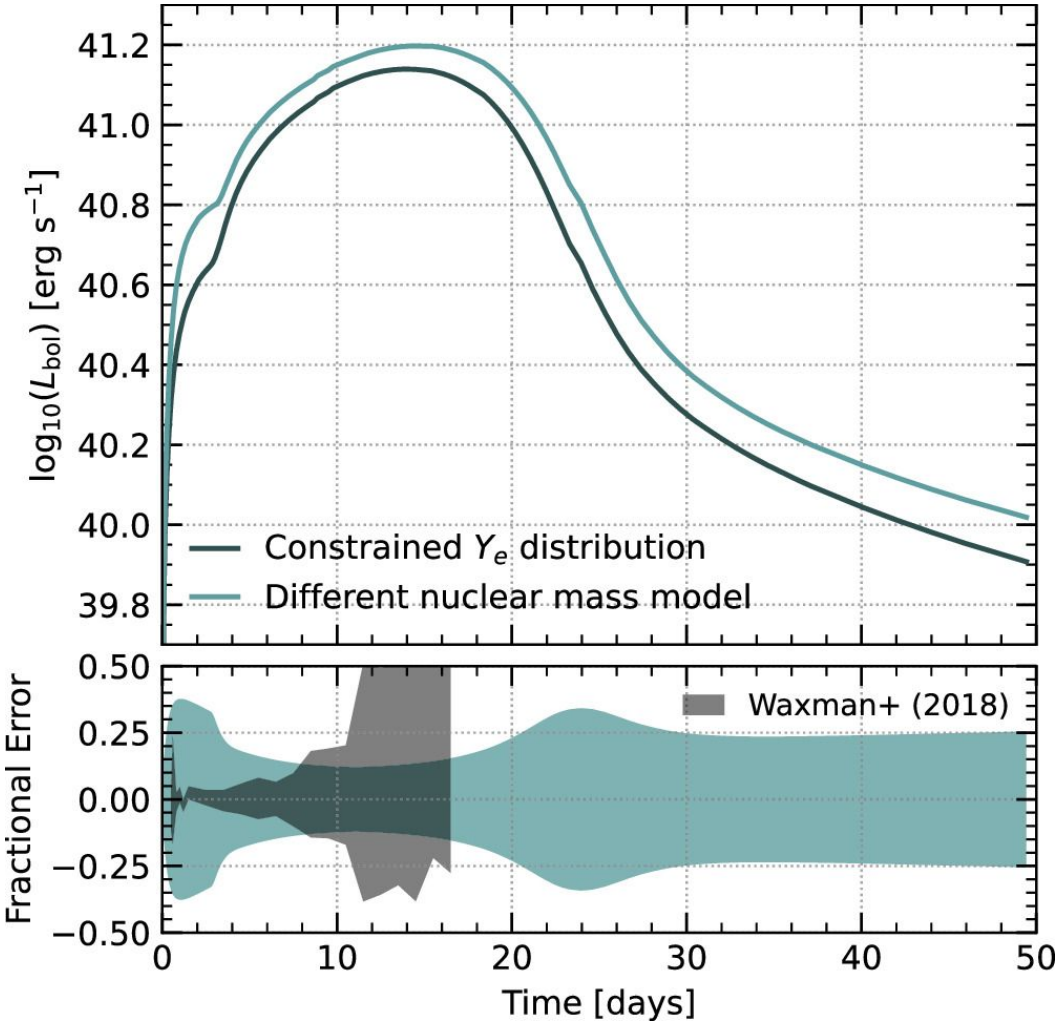
Hubble diagram showing the final results from the HST Key Project (Freedman et al. 2001) In the top panel velocity is plotted versus distance for the five secondary distance methods labeled in the plot. The solid line is for a slope of 72, flanked by dotted lines indicating a 10% uncertainty. The bottom panel shows the residuals for a value of H0 = 72 ± 3 (stat) ±7 (sys) km s −1 Mpc −1 .

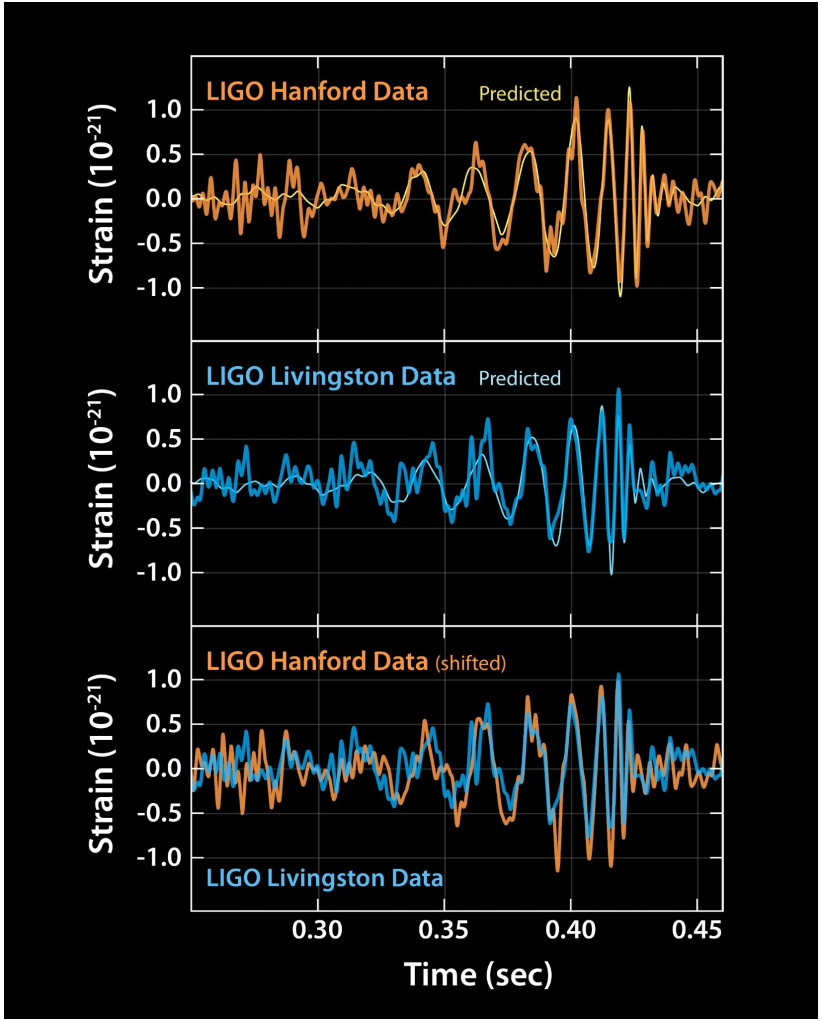

He
C O Ne Mg Si
Fe
AVERAGE BINDING ENERGY per NUCLEON
2
3
4
5
6
7
8
9
Li
Be
B
fusion
fission
0 4 8 12 16 20 24 30 60 90 120 150 180 210 240
MASS NUMBERS A

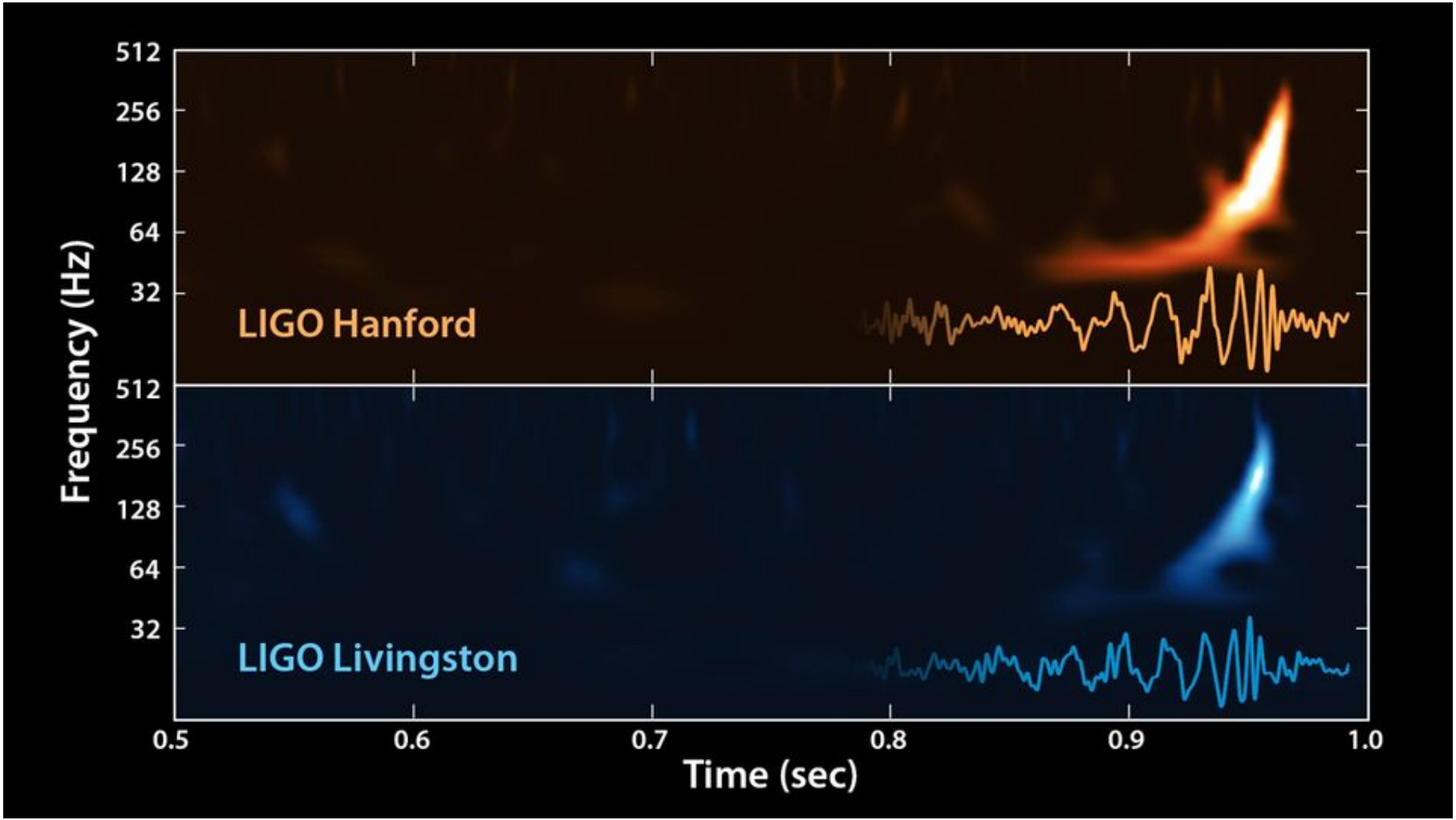

Binding Energy Per Nucleon(MeV)
10
8
6
4
2
Iron..56
0
50
100
150
200
250
Mass Number

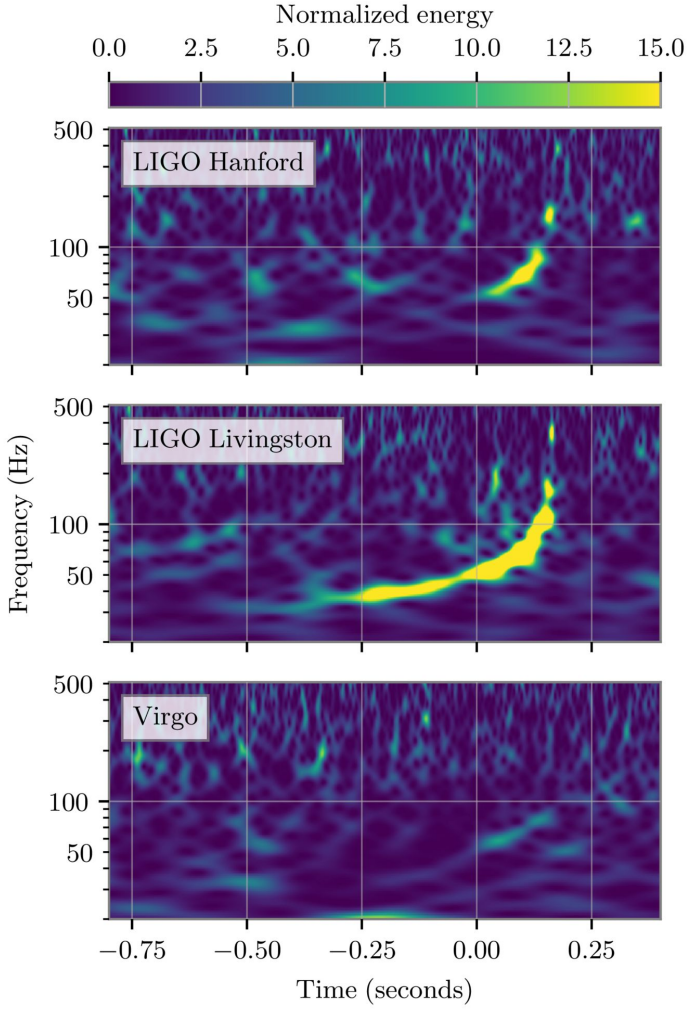
Is This Data??

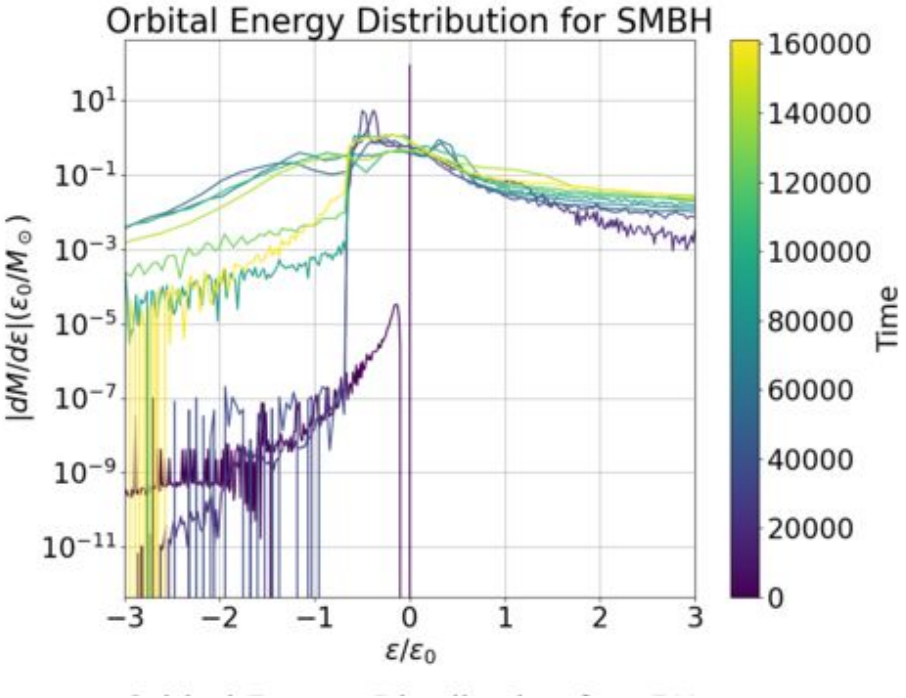

Binding Energy Per Nucleon(MeV)
10
8
6
4
2
Iron..56
0
50
100
150
200
250
Mass Number

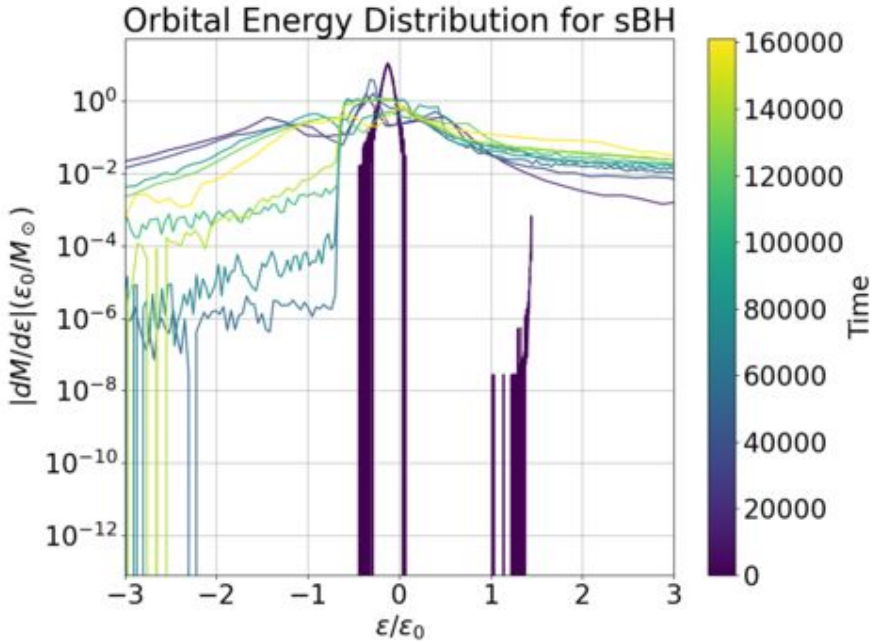

41.2
41.0
40.8
40.6
40.4
40.2
40.0
39.8
$\log_{10}(L_{\rm bol})$ [erg s^{-1}]
Constrained Y_e distribution
Different nuclear mass model
0.50
0.25
0.00
−0.25
−0.50
Fractional Error
Waxman+ (2018)
0
10
20
30
40
50
Time [days]

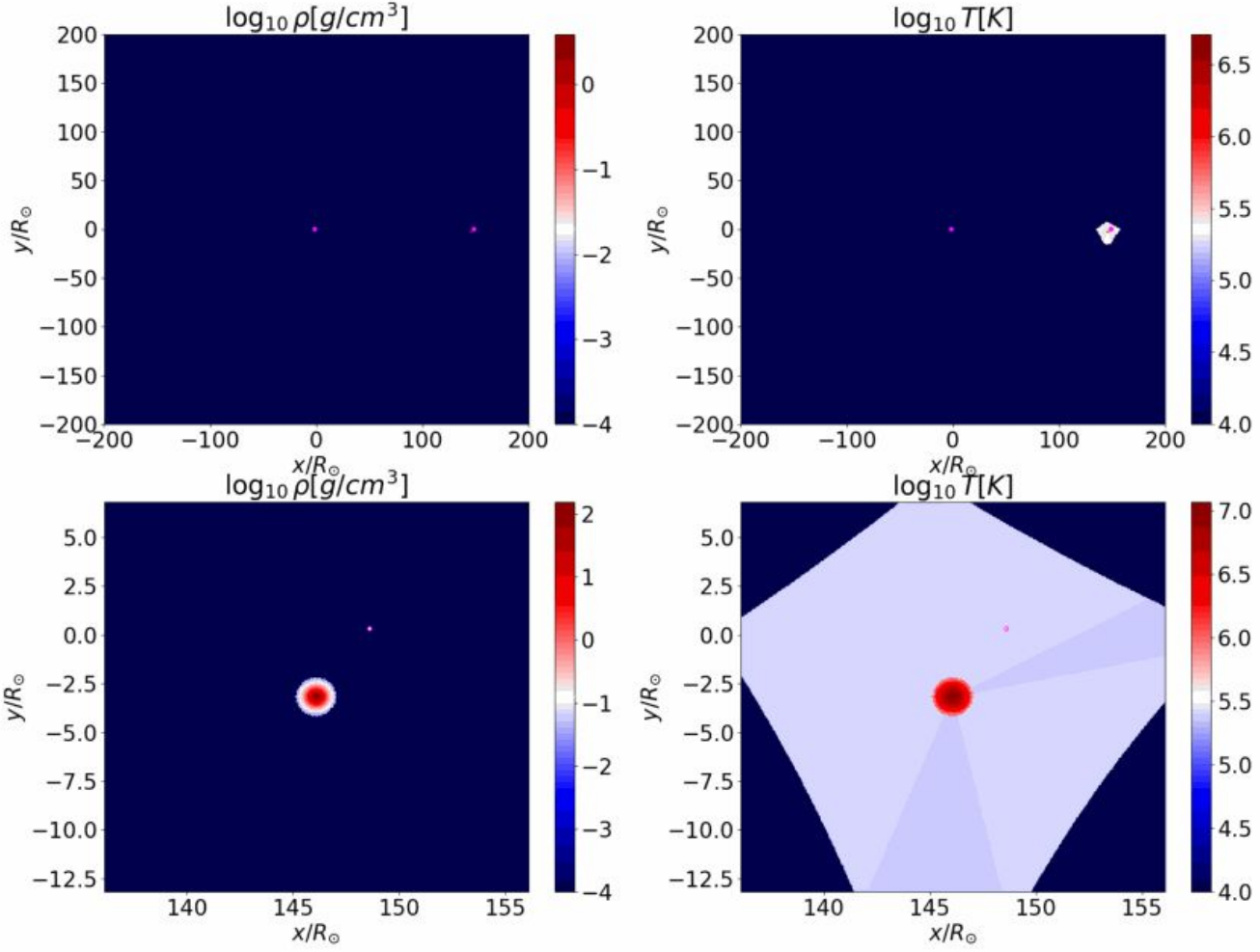

LIGO Hanford Data
Predicted
LIGO Livingston Data
Predicted
LIGO Hanford Data (shifted)
LIGO Livingston Data
Strain (10-21)
1.0
0.5
0.0
-0.5
-1.0
0.30
0.35
0.40
0.45
Time (sec)


512
256
128
64
32
LIGO Hanford
512
256
128
64
32
LIGO Livingston
Frequency (Hz)
0.5
0.6
0.7
0.8
0.9
1.0
Time (sec)


Normalized energy
0.0
2.5
5.0
7.5
10.0
12.5
15.0
LIGO Hanford
LIGO Livingston
Virgo
500
100
50
Frequency (Hz)
−0.75
−0.50
−0.25
0.00
0.25
Time (seconds)


Orbital Energy Distribution for SMBH
$|dM/d\varepsilon|(\varepsilon_0/M_\odot)$
$\varepsilon/\varepsilon_0$
Time
160000
140000
120000
100000
80000
60000
40000
20000
0


Orbital Energy Distribution for sBH
$|dM/d\varepsilon|(\varepsilon_0/M_\odot)$
$\varepsilon/\varepsilon_0$
Time
160000
140000
120000
100000
80000
60000
40000
20000
0


$\log_{10}\rho[g/cm^3]$
$\log_{10}T[K]$
$\log_{10}\rho[g/cm^3]$
$\log_{10}T[K]$
$x/R_\odot$
$y/R_\odot$